Clarifying Without Bias: Chatbots in Experimental Economics

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A standing threat to inference in experimental economics is that subjects may simply not understand the task. When choices reflect confusion about the rules or payoffs rather than underlying preferences, the resulting behavior can mimic the very objects experiments seek to measure: confusion has been shown to inflate apparent fairness, cooperation, trust, and other-regarding behavior across canonical games, and to contaminate elicited risk attitudes. Because the same data are used to build theory and to inform policy, confusion that goes unaddressed distorts both. Experimenters therefore work to clarify instructions, often relying on research assistants to answer subjects' questions during a session—but a human assistant is costly, inconsistent across sessions, and can itself bias choices by hinting at the "right" answer. This motivates our question: can a large language model take over the assistant's role, reducing confusion by clarifying the rules without introducing bias by giving advice?

Answering it requires a chatbot that clarifies but never steers, and a way to measure how reliably it stays within that role. We study the question in Oprea's (2024) experiments on choice under risk, where the interpretation of behavior hinges on separating what subjects *comprehend* from how *complex* the task is to evaluate. Our chatbot may restate rules and define terms, but is forbidden to give advice, compute the best option, or reduce the task's complexity. To evaluate it without running human sessions, we have adversarial "synthetic subjects"—language models instructed to behave like difficult participants who seek advice, demand worked examples, or probe for the experimenter's intent—attack the chatbot over many fifteen-turn conversations, more than 27,000 replies in total, while an independent model scores every reply for advice-giving, factual error, and incoherence. Within this design we compare alternative chatbot architectures and base models.

Two results stand out. First, a *simpler* design is more reliable: a single, well-instructed model call adheres best, and each additional "guardrail" agent we add makes advice-giving more common, not less, because the extra steps tend to repeat the subject's forbidden request back to them. Second, the one change that clearly helps is a *stronger base model*, which cut rule-breaking replies from about 12% to 4% on the hardest case while also improving accuracy and speed; adding reasoning steps or a correction pass did not help. Because small per-reply risks accumulate, we evaluate reliability over a whole conversation rather than a single turn. These findings are preliminary—a confirmation that fully separates the chatbot from the model judging it is still running—but they already suggest that, for a chatbot meant to clarify without biasing, the capability of the base model matters more than the elaborateness of the surrounding workflow. Full methods and interactive results: `can-celebi.github.io/chatbot-for-experiments`.